

Set & Logic Notation Reference

CSCE 763 — J.C. Vaught

1 Set Operations (use these on the exam)

Symbol	Name	Meaning	Plain English
$\cap$	Intersection	$A \cap B$: elements in both	AND
$\cup$	Union	$A \cup B$: elements in either	OR
$A - B$	Difference	in A but not B	A AND NOT B
A^c	Complement	not in A	NOT
$\subseteq$	Subset	every element of left is in right	“fits inside”
$\subset$	Strict subset	subset and not equal	“fits, smaller”
$\in$	Element of	$x \in A$	“x is in A”
$\notin$	Not element of	$x \notin A$	“x is not in A”
$\emptyset$	Empty set	no elements	“nothing”
$ A $	Cardinality	number of elements	“size of A”

2 Morphology Operators

Symbol	Name	Definition
$\hat{B}$	Reflection	$\{w \mid w = -b, b \in B\}$
$(B)_z$	Translation	$\{c \mid c = b + z, b \in B\}$
$\ominus$	Erosion	$A \ominus B = \{z \mid (B)_z \subseteq A\}$
$\oplus$	Dilation	$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$
$\circ$	Opening	$A \circ B = (A \ominus B) \oplus B$
$\bullet$	Closing	$A \bullet B = (A \oplus B) \ominus B$
$\circledast$	Hit-or-miss	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$

3 Pure Logic (propositions, not sets)

Symbol	Name	Meaning
$\wedge$	AND	both true
$\vee$	OR	at least one true
$\neg$ or $\sim$	NOT	flip truth
$\Rightarrow$	Implies	if-then
$\Leftrightarrow$	If and only if	both directions
$\forall$	For all	every
$\exists$	There exists	at least one

Key distinction. $\wedge/\vee$ work on *propositions* (true/false statements). $\cap/\cup$ work on *sets*. They mean the same thing logically but appear in different contexts. On a morphology or segmentation exam, you almost always use $\cap$ and $\cup$.

4 How These Show Up in Course Topics

4.1 Region Segmentation

- Disjoint regions: $R_i \cap R_j = \emptyset$ for $i \neq j$ (no overlap).
- Cover the image: $\bigcup_{i=1}^n R_i = R$ (every pixel is in some region).
- Merge predicate: if $Q(R_i \cup R_j) = \text{TRUE}$, merge R_i and R_j .
- Each region must be connected: every pair of pixels in R_i has a path entirely inside R_i .

4.2 Connectivity / Components

- Pixels p, q are connected in S if a path of pixels in S joins them.
- Connected component $C \subseteq S$: maximal connected subset.
- $S = C_1 \cup C_2 \cup \dots \cup C_k$ with $C_i \cap C_j = \emptyset$ for $i \neq j$.

4.3 Morphology Examples

- Erosion “fits”: SE must fit entirely inside A .
- Dilation “hits”: reflected SE must overlap A at least once.
- Boundary: $\beta(A) = A - (A \ominus B)$.
- Duality: $A \oplus B = (A^c \ominus \hat{B})^c$.
- Idempotence: $(A \circ B) \circ B = A \circ B$, $(A \bullet B) \bullet B = A \bullet B$.

5 Translation Quick-Test

Expression	Plain English
$A \cap B = \emptyset$	A and B do not overlap
$A \subseteq B$	A fits inside B
$p \in (A \cap B)$	pixel p is in both regions
$A \cup B$	combine A and B into one set
A^c	the background (everything not in A)
$A - B$	A with B removed
$ A > 0$	A is nonempty
$\forall p \in A : f(p) > T$	every pixel in A has intensity above T
$\exists p \in R : Q(p) = \text{TRUE}$	some pixel in R satisfies Q

6 Five-Symbol Legend (worth putting on cheat sheet)

$\cap$ = AND / intersection $\cup$ = OR / union $\subseteq$ = subset A^c = complement $\emptyset$ = empty set

That covers essentially every set-theory symbol the final exam can throw at you.